Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

July 27, 2026

Dear Respondent,

Greetings!

We are fourth-year Bachelor of Science in Aviation Information Technology (BSAIT) Section 2 students from the National Aviation Academy of the Philippines, conducting our capstone project entitled: **"A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"**

As part of the system evaluation, we respectfully invite you to evaluate the developed system. Your honest responses will help determine the quality and acceptability of the system and will serve as valuable input for improving its functionality and overall performance. Your participation is voluntary. You may decline to answer any question or withdraw from the evaluation at any time without any penalty.

Respectfully,

Charmaine Rose G. Angeles
Researcher

Kylane C. Gravino
Researcher

Melody D. Sedanto
Researcher

Noted by:

RICHARD NIEL B. PERALTA, MSIT
Capstone Adviser

ASSOC. PROF. MARY ANN ABALLIAR-VISTA, MEAM
Academic Adviser

In accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), all information collected through this questionnaire will be treated with the utmost confidentiality and will be used solely for academic and research purposes. Your identity will remain anonymous, and the information you provide will not be disclosed to any unauthorized person or organization. The collected data will be securely stored and disposed of after the study's completion.

By completing this questionnaire, you indicate your voluntary consent to participate in this study.

Consent Declaration

☐ I have read and understood the Data Privacy and Confidentiality Statement.

☐ I voluntarily agree to participate in this research study.

Thank you for your valuable time and participation.

Capstone Title: "A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Conduct a comparative analysis between the existing manual paper-based system and the developed centralized digital job order entry and management system in terms of service record accessibility, retrieval time, and accuracy.
2. Evaluate the machine learning model for predicting service turnaround duration and corresponding release dates using historical service data in terms of its predictive performance based on appropriate model evaluation metrics.
3. Determine the impact of the real-time analytics in the Sales Report page on reporting efficiency by assessing the time required to generate sales, expense, and return on investment (ROI) reports.
4. Evaluate the effectiveness of the developed system based on the ISO/IEC 25010 software quality standards, specifically in terms of:
    - 4.1 Functional suitability;
    - 4.2 Performance efficiency;
    - 4.3 Compatibility;
    - 4.4 Interaction Capability;
    - 4.5 Reliability;
    - 4.6 Security;
    - 4.7 Maintainability;
    - 4.8 Flexibility; and

4.9 Safety;

5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

Survey Questionnaire

Part I. Respondent Profile

Instructions: Please provide the following information by filling in the data or putting a (✓) next to your response.

Name: (Optional) Ken Carlo Javier

Role: ☐ IT Expert ☑ IT Practitioner

Position/Job Title: Software Engineer Years of IT Experience:

☑ 5–7 years ☐ 8–10 years ☐ More than 10 years

Part II. System Evaluation (ISO 25010)

Instructions: To indicate your agreement level that best describes how you evaluate the system based on ISO 25010 software quality characteristics, put a (✓) in the box that is next to each statement below using the coded scale for your response.

Scale	Interpretation	Description
4	**Highly Acceptable**	The system fully meets the evaluation criterion and performs exceptionally well.
3	**Acceptable**	The system meets the evaluation criterion and performs satisfactorily.
2	**Less Acceptable**	The system partially meets the evaluation criterion and requires improvement.
1	**Not Acceptable**	The system does not meet the evaluation criterion and requires significant improvement.

CRITERIA	4	3	2	1
Functional Suitability- *This section evaluates whether the system provides the necessary functions to perform daily tasks.*				

CRITERIA	4	3	2	1
1. The system provides the necessary functions for managing job orders, customer information, inventory, reports, and release schedules.	✓			
2. Each system module produces the expected result based on the user's operation.	✓			
3. The Machine Learning component uses a Random Forest Regressor to analyze job order data, identify service patterns, and generate a predicted release date for operational decision-making.	✓			
4. The system's data analytics functions provide information for monitoring sales, services, and business performance.	✓			
5. The system supports the complete job order workflow, including New Order, On-Going, For Release, and Claimed statuses.	✓			
Performance Efficiency- *This section evaluates whether the system performs tasks quickly and efficiently.*				
1. The system retrieves database records and loads pages within an acceptable response time.	✓			
2. Job order processing and saving transactions are completed without noticeable delays.	✓			
3. Sales and expense reports can be generated for daily, weekly, monthly, quarterly, and annual periods within an acceptable response time.	✓			
4. The Machine Learning component generates a predicted release date from valid input data without noticeably delaying the transaction workflow.	✓			
5. Navigation between system modules occurs without noticeable delay, supporting efficient single-page application (SPA) routing.	✓			
Compatibility- *This section evaluates whether the system works properly with other system components and supported platforms.*				
1. The system operates properly and maintains its functional layout across commonly used web browsers.	✓			
2. Modifications made to existing records, such as job order status changes, are immediately reflected in all related system modules.	✓			
3. Information entered in one module is correctly synchronized across other system modules.	✓			
4. The Machine Learning output is correctly integrated into the Job Order module to display the predicted release date.	✓			
5. The system's data analytics functions retrieve and consolidate transaction data to display current operational metrics.	✓			
CRITERIA	**4**	**3**	**2**	**1**
Interaction Capability- *This section evaluates whether the system is easy to learn, understand, and use.*				

CRITERIA	4	3	2	1
1. The system's menus, icons, and buttons are clearly labeled and easy to recognize.		✓		
2. The system's job order, inventory, service, and user management features are easy to learn.	✓			
3. The system provides validation messages and confirmation prompts to help prevent input errors and unintended actions.	✓			
4. The layout and navigation make it easy to move between different modules.	✓			
5. Frequently used tasks can be completed with minimal steps.	✓			
Reliability- *This section evaluates whether the system performs consistently and operates without failures.*				
1. The system performs consistently without unexpected interruptions during operation.	✓			
2. The system handles invalid or incomplete input without unexpected failure.	✓			
3. Transactions are processed without data loss or unintended changes to existing records.	✓			
4. The system supports manual PostgreSQL database backups through the Heroku platform to facilitate data recovery.	✓			
5. The Machine Learning component consistently generates a predicted release date when provided with a complete and valid job order input.	✓			
Security- *This section evaluates whether the system protects data from unauthorized access.*				
1. The system uses JSON Web Token (JWT) authentication with bcrypt password hashing, enforces a three-attempt login limit with a 15-minute lockout, and automatically terminates inactive sessions after 30 minutes, addressing OWASP A07: Identification and Authentication Failures.	✓			
2. The system enforces Role-Based Access Control (RBAC) for Staff and Owner roles and prevents direct access to unauthorized modules through URL requests, addressing OWASP A01: Broken Access Control.	✓			
3. The system uses parameterized queries and input validation to prevent SQL injection and invalid data submission, addressing OWASP A03: Injection.			✓	
4. The system records critical user activities and system events in audit logs, including actions, user information, timestamps, and relevant changes, addressing OWASP A09: Security Logging and Monitoring Failures.	✓			
5. The system uses HTTPS with SSL/TLS encryption to protect data transmitted between the client and application server, addressing OWASP A02: Cryptographic Failures.	✓			

CRITERIA	4	3	2	1
Maintainability- *This section evaluates whether the system is easy to maintain, update, and improve.*				
1. The system provides clear error messages identifying the specific field and cause of invalid input to assist troubleshooting.	✓			
2. The system allows incorrect data entries to be corrected without affecting completed transaction records.	✓			
3. The system remains functional when existing records, services, or configurations are modified.	✓			
4. The system logs administrative actions and critical errors to facilitate system monitoring and maintenance.	✓			
5. The system's modules can be updated or modified without disrupting the job order workflow.	✓			
Flexibility- *This section evaluates whether the system can adapt to future changes and business needs.*				
1. The system supports different user roles, such as Staff and Owner, with appropriate access permissions and functions.	✓			
2. The system supports different types of shoe services within the same workflow.	✓			
3. The system handles different customer transactions, such as single-service and multi-service orders within the same workflow.	✓			
4. The system accommodates different payment methods and payment statuses within the same transaction workflow.	✓			
5. The system allows new services to be added without affecting existing transactions.	✓			
Safety*- This section evaluates whether the system helps prevent errors and protects important data.*				
1. The system helps prevent accidental loss or deletion of important records.	✓			
2. The system requires explicit user confirmation before executing critical data deletions or modifications.	✓			
3. The system enforces mandatory field requirements before allowing a transaction to proceed.			✓	
4. The system aborts transactions that violate defined business logic or validation rules.			✓	
5. The system prevents duplicate submissions during transaction processing.	✓			

PART III. Comments and Suggestions

Instructions: Please provide your comments and recommendations regarding the developed system.

1. **Which feature or component of the system do you think performs best? Why?**

The Machine Learning integration for predicting release dates. It calculates predictions synchronously (~1.2s) during order creation without blocking the UI, adding immediate, practical value to the workflow.

2. **What weaknesses or issues did you observe during your evaluation?**

Backend API validation is overly permissive. While the UI correctly blocks empty fields, direct API testing shows it accepts empty or malformed payloads and assigns default values (e.g., 0.00 for totals) instead of rejecting them, lacking defense-in-depth. See **ORD-2026-09-19-001** for example.

3. **What improvements do you recommend to enhance the system's quality in terms of functionality, performance, security, maintainability, or other software quality characteristics?**

Enforce strict backend validation to reject incomplete payloads at the API level. Implement Redis caching for the Sales/Analytics modules to reduce database load as data scales, and add rate-limiting to the login endpoint to prevent brute-force attacks.

4. **Are there additional features or enhancements that you would recommend for future versions of the system?**

- **UI/UX & Accessibility:** Enhance the interface to comply with WCAG standards. Improve color contrast (the current gray text is difficult to read) and ensure proper font sizing for better readability.
- **Notifications:** Automated SMS/Email notifications for customers when their order status changes.
- **Inventory:** Auto-deduction of raw materials from inventory when a job order status changes to “On-Going”.
- **Machine Learning:** An automated pipeline to retrain the ML model weekly with new historical data.

5. **Other comments or suggestions:**

Excellent execution. The tech stack (FastAPI, React/Vite, PostgreSQL) is modern and robust. It is evident that AI/LLMs were utilized to assist in development, as the UI and structural design follow common AI-generated patterns. Nevertheless, the team successfully customized these foundations and bridged standard CRUD operations with practical machine learning, resulting in a highly scalable and professional business tool. Great capstone work.

"We assure you that your responses will be treated with the utmost confidentiality and used strictly for academic purposes."

Signature of the Respondent

Date: September 19, 2026